Use Case 1: Producer Product Listing and Inventory Management

A registered producer wants to list a new product on the marketplace. The producer submits the required product information, including product name, description, category, price, available quantity, availability period, farm origin and any relevant information such as harvest date, allergen details, organic certification and best-before date.

The system validates the submitted information. If mandatory information is missing or invalid, the producer is asked to correct the details. If organic certification is claimed, the certification information should be checked using an external or simulated verification service.

Once all required information has been successfully validated, the product is published and becomes available to customers. The producer can later update the product quantity and seasonal availability.

When stock reaches zero, the product should automatically become unavailable for new orders. If stock is replenished and the product is still within its seasonal availability period, it can become available again.

The flow should also consider what happens if the external verification service is unavailable or validation fails.

Use Case 2: Customer Multi-Vendor Order, Checkout and Payment

A customer browses or searches the marketplace and adds products from one or more producers to their basket.

When the customer proceeds to checkout, the system checks that all selected products are still available in the requested quantities. If an item is unavailable or there is insufficient stock, the customer is informed and asked to amend the order.

If the products are available, the system identifies the producers involved and determines the relevant delivery or collection arrangements. The customer reviews the order and confirms the purchase.

The system then creates the overall marketplace order and separates it into the relevant producer-specific order elements.

Payment is submitted to an external or simulated payment service. If payment is successful, the order is confirmed, the appropriate inventory quantities are updated and the relevant producers are notified.

If payment is declined, the customer is informed and given the opportunity to retry. If the payment service is temporarily unavailable, the process should follow an appropriate exception or retry path rather than simply terminating.

For multi-vendor orders, the system should calculate the amount attributable to each producer and calculate the marketplace's 5% commission.

For assessment purposes, students should use a sandbox or simulated payment service rather than a live payment system.

Use Case 3: Food Quality Incident and Customer Traceability

A customer, producer or marketplace administrator reports a potential food-quality or safety issue relating to a product sold through the marketplace. The issue could involve an undeclared allergen, incorrect best-before information, contamination or another product-quality concern.

An incident record is created and the relevant product is identified. The system retrieves the producer information and determines which customer orders contained the affected product.

The responsible producer is contacted and asked to provide further information about the issue. Based on the information received, the incident may be closed if no genuine risk is identified or escalated if customers could be affected.

If the issue is confirmed, the product should be temporarily suspended from further sale where appropriate.

The system identifies all affected customers and sends them an appropriate notification. The result of each notification should be recorded. If a notification fails, the process should identify this so that an alternative communication method or manual follow-up can take place.

Once all required actions have been completed, an authorised marketplace representative reviews the incident and either closes it or requests further action.

The base case explicitly requires records linking customers, products and producers so that rapid communication can take place when quality or food-safety concerns arise.